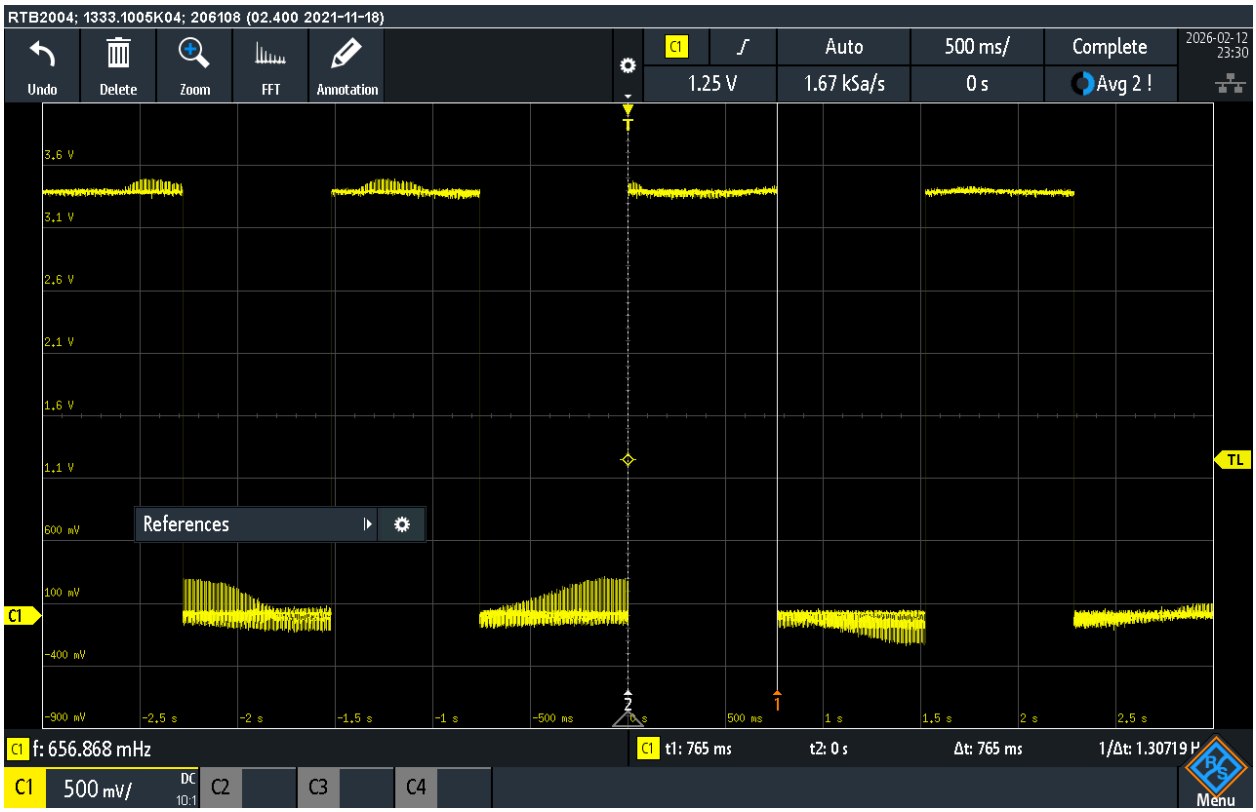
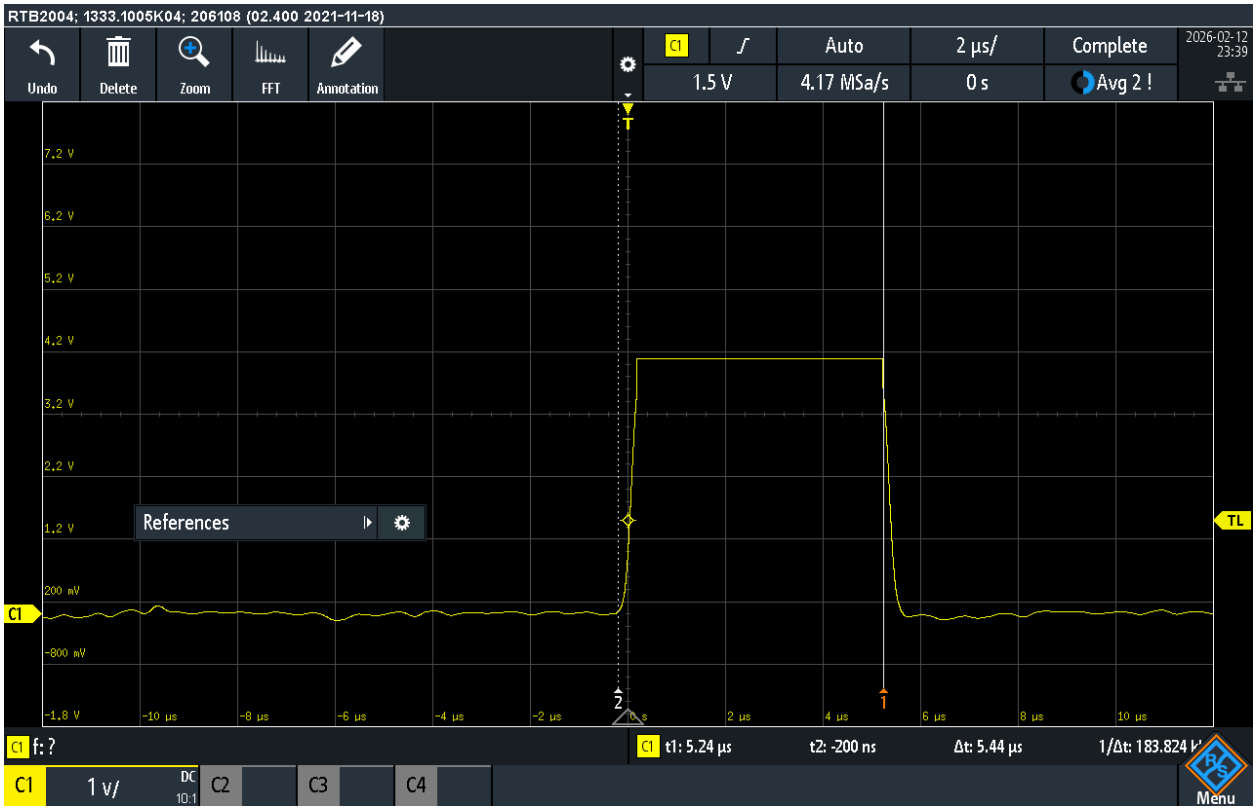
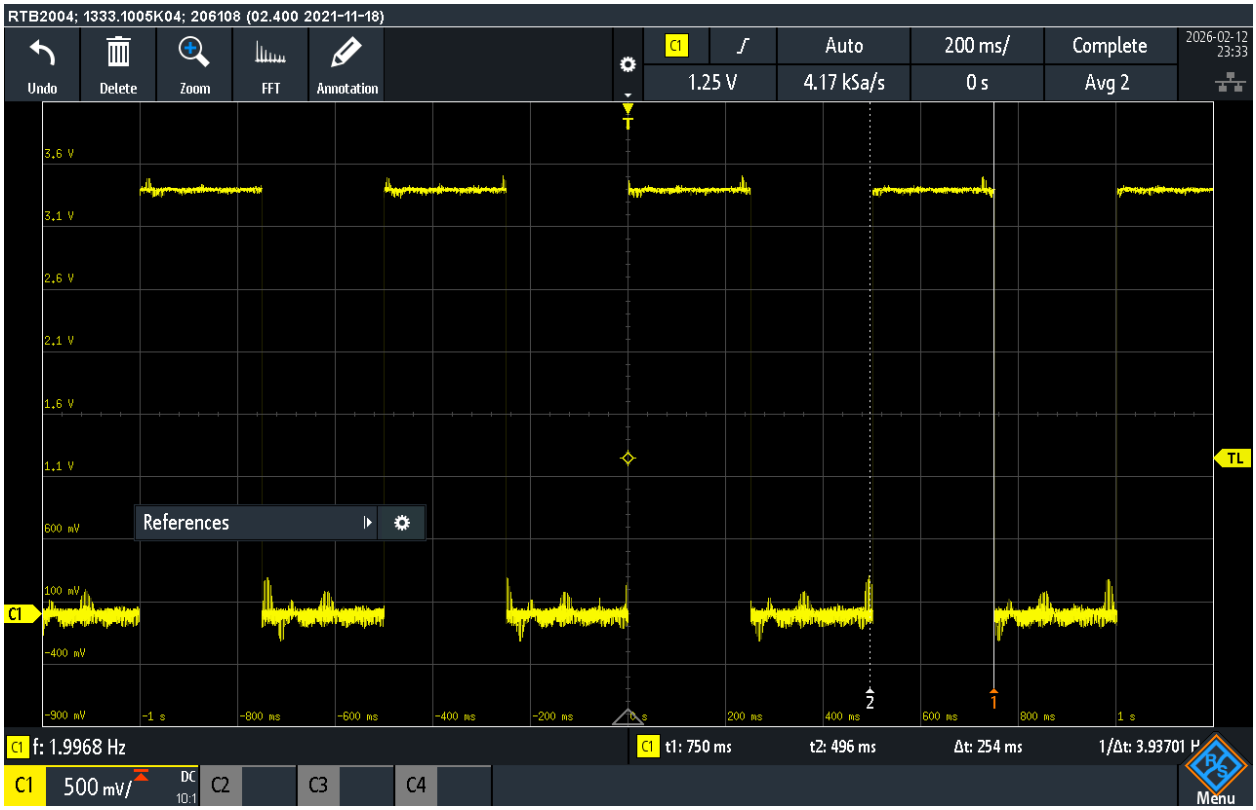


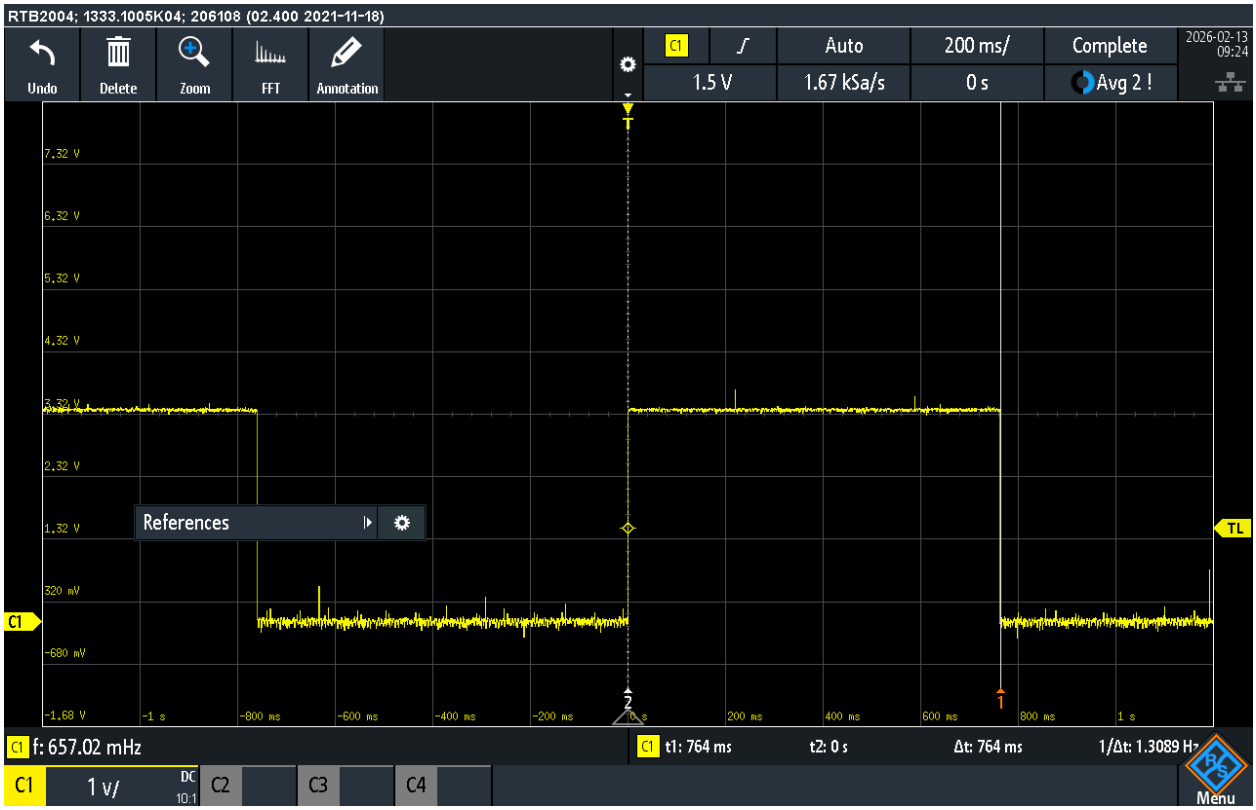
A measurement of pre debug latch code executing the 0.66 Hz blink. Actual timing is 656.868 mHz -> 0.476808125% error, within specified tolerance.



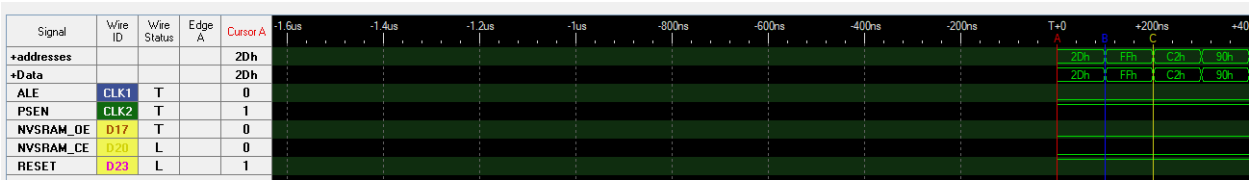
A measurement of how a fast a previous iteration of the ISR code executed in. 5.44 milliseconds, below the expected time.



A measurement of the fast blink, enabled by pulling pin 1.3 to Vcc. Approximately 2.0Hz (no tolerance specified, so percent error calculations not done).



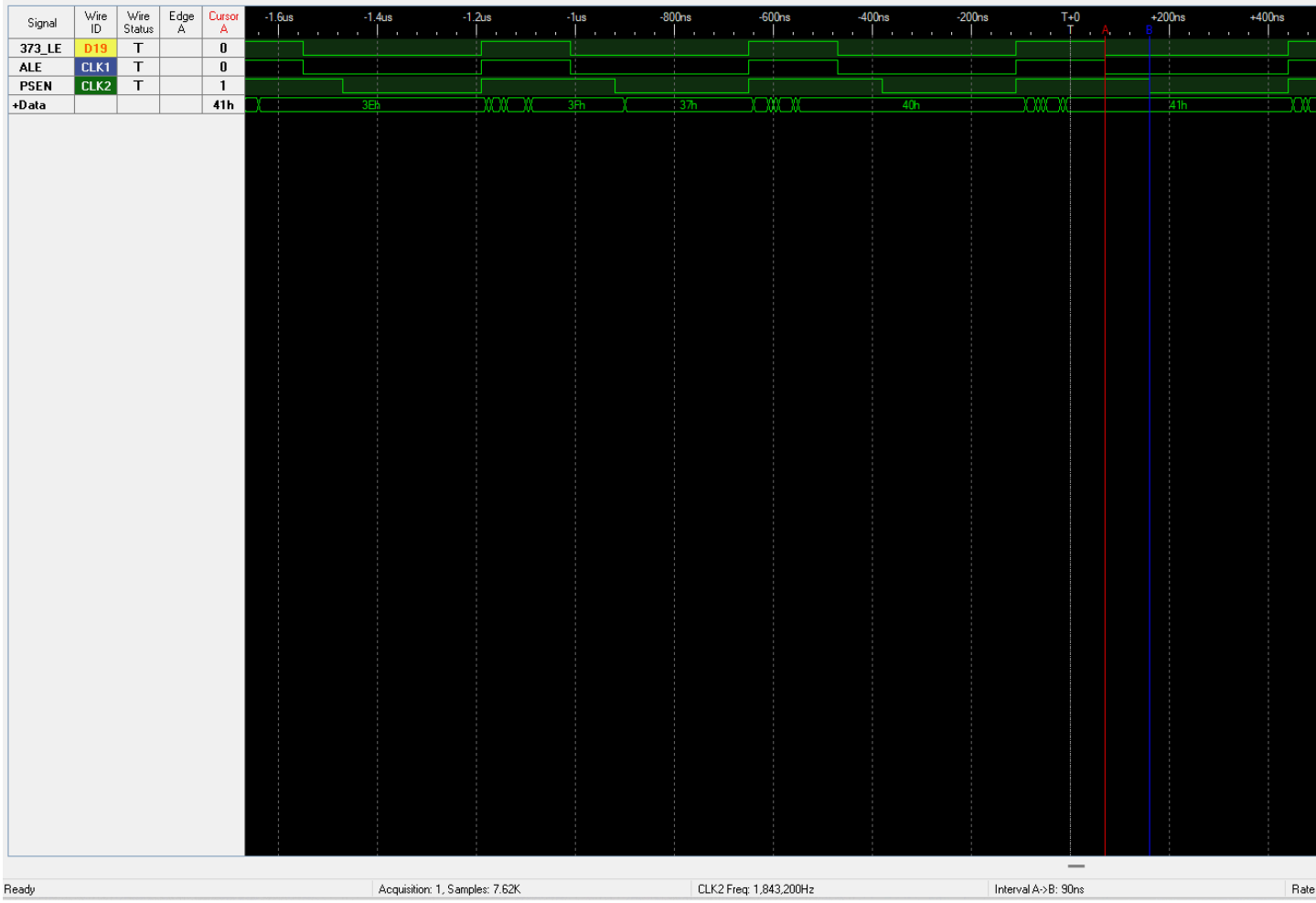
A measurement of the post debug latch code implementation timing of the 0.66Hz blink code. 0.453563057% error.



A statemode capture of instructions on the Logic analyzer
Setup - Capture the rising edge of PSEN, so that the NVSRAM is still emitting the instruction.
Trigger is getting the second byte of the first instruction

The second instruction byte of the first instruction is shown because we are capturing rising edge of PSEN. PSEN goes low for the first time on the first byte.

We Capture 2D, (second byte of first instruction, or immediately after A), FF (what follows SJMP, immediately after B), C290 (the next instruction actually executed, immediately after C), so on and so forth



A screenshot of timing mode capturing the information needed to determine if the datasheet is obeying t_{llpl}. So in this screenshot, A is when the latch enable/ALE signal goes low, and B is when the PSEN goes low. Difference of 90ns between the edges.

For the AT89c51rc2 8051, the Tllpl / Time From ALE low to PSEN Low minimum is calculated by T - x , where T is the oscillator clock period (90.422 ns) and x is 15 ns (given by datasheet). Calculated to be a minimum of 70.422ns. Measured value is 90ns. So the 8051 is obeying the datasheet

So for the 373 latch, the hold time (time the signal must be stable after the falling edge) is 20ns. Which means the latch must be stable in 70 ns, which it is

Timing Analysis

So the setup time for ‘374 latch is 20 ns and the hold time is 0ns. A is currently at +100ns, and /WR toggles at B, which is +200ns. So the signal is stable for at least 100ns before hand, which is greater than the setup time.

Signoff Questions

1) What operating system (including revision) did you use for your 8051 code development?
Windows 11 24H2 Enterprise - Version of windows on the lab computers
2) What assembler(s) (including revision) did you use?
Keil uVision V5.43.1.0 - Version of keil on the lab computer
3) Did you install and use any other software tools to complete your lab assignment?
Used FLIP and LogicPort, but those were already installed.
4) Did you experience any problems or challenges with this lab assignment or any of the software tools? If so, describe the issues.
Nothing beyond usual debugging issues.
5) If you have any suggestions for changes to this lab assignment for the future, please include those ideas in your submission.
No suggestions at this time.